Lohit Gali

Systems Software Engineer

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Professional Experience

Systems Software Engineer *Jun 2025 – Present*

Epik Solutions | Enterprise Systems Software

- Led design, implementation, and deployment of **Hardware and OS abstraction** modules in **C++** for enterprise software, improving compatibility and reducing integration time by 25%.
- Developed **Linux system stack** components in **C++** to monitor reliability and error handling across telemetry pipelines, preventing failures and boosting processing uptime by 15%.
- Optimized **system infrastructure services** by refactoring low-level **I/O** and **memory management** paths, achieving 30% lower latency for security data ingestion workloads across production environments.
- Architected scalable **distributed** data processing backends in **Java**, ensuring continuous operation, integrity, and resilience under peak loads for complex customer environments and enterprise services.
- Performed **kernel debugging** and diagnostic analysis on **Linux modules**, resolving resource contention and crashes while improving production system stability by 20% across customer environments.
- Implemented automated test frameworks and **CI/CD pipelines** for **systems programming** components, improving code quality and enabling reliable enterprise software updates with minimal downtime risk.

Software Engineer *Jul 2024 – Jun 2025*

Samay Systems LLC | Cloud Compliance Systems

- Engineered high-performance **system infrastructure services** in **Java** to collect evidence data from **AWS**, **Azure**, and **GitHub APIs** for multi-tenant compliance validation at enterprise scale.
- Designed optimized **data structures and algorithms** for processing compliance metadata efficiently, reducing large dataset processing time by 40% across audit workflows and reporting pipelines.
- Contributed to scalable **distributed system** design for real-time compliance checks, improving audit data reliability by 35% for enterprise clients across production governance workflows reliably.
- Developed robust **API integrations** in **C++** for cloud platforms, abstracting complexity for downstream services and standardizing compliance engine inputs across AWS and Azure environments.
- Collaborated on complex **relational schemas** and **SQL workflows**, enabling data protection technologies and auditable data management for multi-framework compliance programs across enterprise customer deployments.
- Deployed scalable services using **AWS** infrastructure, **Docker**, and **Kubernetes**, establishing reliable deployment practices and high availability across multiple production environments under customer workloads continuously.

Software Engineering Intern *Jun 2023 – Aug 2023*

Monarch Tractor | Embedded Linux Systems

- Supported **systems programming** components in **Python** and **C++** for processing sensor and image data from semi-autonomous agricultural vehicle platforms during engineering validation and testing.
- Developed high-performance preprocessing pipelines transforming raw sensor data into training-ready datasets, enhancing throughput by 20% for perception and prediction tasks on vehicle data streams.
- Assisted in optimizing **Linux stack** configurations and drivers for improved embedded data acquisition performance, ensuring reliable streams for vehicle engineers during field test cycles.
- Performed **kernel debugging** on embedded **Linux** systems, diagnosing driver and resource allocation issues while improving stability of autonomous vehicle operations during platform validation cycles.

Junior Software Engineer *Jul 2021 – Aug 2022*

ZeMoSo Technologies | Backend Software Services

- Contributed to backend software services in **Java** for a cloud-based scientific operations platform, improving experiment and sample data traceability across laboratory workflow modules reliably.
- Developed scalable data validation and enrichment pipelines using **Python** and **SQL**, ensuring high quality enterprise data consistency for scientific teams and reporting workflows daily.
- Applied **data structures and algorithms** to optimize processing workflows, reducing experiment data preparation time by 25% and enabling downstream modeling for scientific analytics teams.
- Integrated **CI/CD pipelines** to automate testing and deployment of software services, enforcing quality checks and maintaining governance auditability across releases for regulated platform updates.

Skills

Programming Languages: C++, Java, Scala, Python, Bash, PowerShell
Operating Systems & Kernel: Linux, Linux Kernel, Kernel Debugging, OS Internals, System Calls
Systems & Distributed Technologies: Data Structures, Algorithms, Distributed Systems Design, High Performance Computing, Concurrency, IPC, Networking, API Design
Storage & Data Protection: Storage, Filesystems, Data Protection, Block Storage, Deduplication, Replication
Tools & Platforms: Docker, Kubernetes, Git, Jenkins, GitHub Actions, Azure DevOps, Terraform, AWS, Azure

Education

Master's in Computer Science, Oklahoma State University, Stillwater, OK
Bachelor's in Computer Science, JNTU Hyderabad, India